

CHAPTER 02: Mobile Communication Technologies

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CH02: Topics to be covered

- Overview on Mobile Technologies,
- Current Mobile technologies (GSM, GPRS, UMTS, LTE, 5G)
- Mobile Cloud Computing
- Recent related Technologies

- What is Telecom Network ?

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 - Telecommunications

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- **Telecommunications** are the means of electronic transmission of information over distances. $\mapsto$ **TELECOM NETWORKS**



is an arrangement of computing and telecommunications resources for communication of information between distant locations.

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2. **Devices** $\mapsto$ that process information and are interconnected by the network.
3. **Telecommunications links** $\mapsto$ Transmission channel
4. **Telecommunications equipment** $\mapsto$ that facilitates the transmission of information.
5. **Telecommunications software** $\mapsto$ that controls message transmission over the network.

- The networks are based either on **wired** or **wireless infrastructures**



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Backbone for Mobile Communication

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 - is technologies w/c allows the exchange of **information** using a communication infrastructure at the same time **regardless of any physical link**
 - Mobile Computing Vs Mobile Communication $\mapsto$ **are same ?**

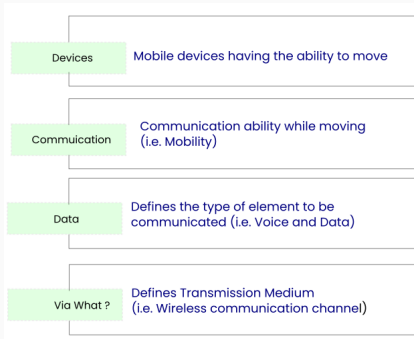
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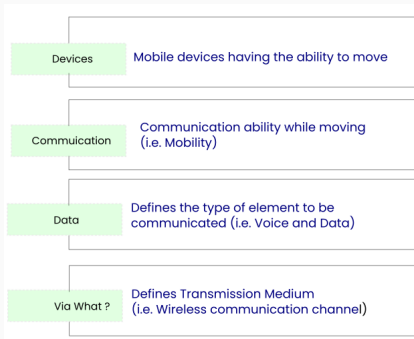


"the ability to use the mobile technology while on the move, most of the portable computers and computing equipment which are particular for the use in stationary place or configuration"

CH02: Mobile Communication Technologies



CH02: Mobile Communication Technologies

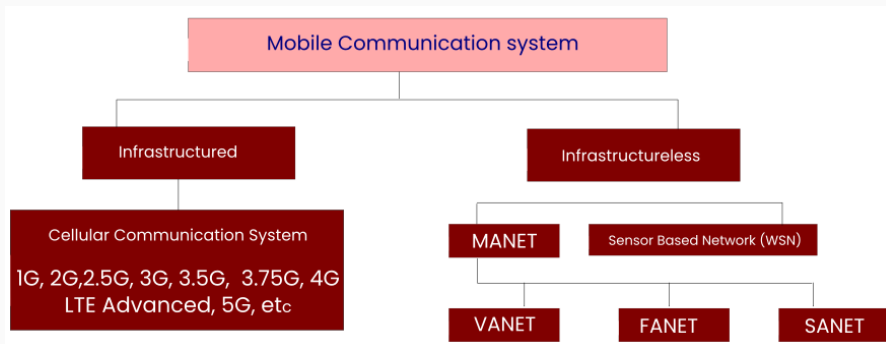


- **Why we need Wireless Communication?**
 - No cost of installation
 - Quick Communication without prolonged setup
 - Serve Large population without wires
 - ubiquitous connectivity

- How to categorize Mobile communication system ?

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- **PSTN (Public-switched telephone network)**
 - is a Global network over worldwide!
- **Channel** : Various types of channels
 - **Control Channel** :
 - is radio channel used for transmission of call request, call setup, call initiation and other beacon
 - **Page** : the actual message which is broadcast over the entire service area

Terminologies ...

- Channel ...
 - Forward Control Channel
 - Radio channel used for transmission of information from Base station to mobile station
 - Reverse Control Channel
 - Radio channel used for transmission of information from mobile station to base station
- Handoff
 - The process of transferring a mobile station from one channel or base station to another
- simplex, Half-duplex and Full-duplex Check

0-Generation Telecom/Mobile Communication tech

1. Mobile Telephony System

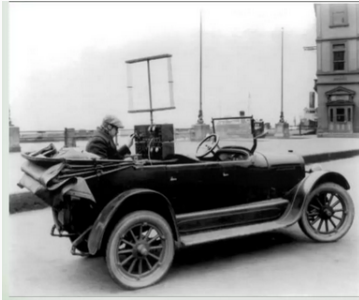
- The first public mobile telephone system → Mobile Telephone System (MTS)
- It was also known as **Car-based mobile Telephony**
 - **Mobile transceivers** of MTS were very big and could be carried only by vehicles.
- MTS was Analog system
 - meaning that it processed voice information as a continuous waveform.
- The system was half-duplex
 - Communication parties @ a time could either speak or listen.
 - To switch between the two modes, users had to push a specific button on the terminal.
- MTS utilized a Base Station (BS) with a single high-power transmitter.
- Base Stations utilized the same frequencies.
- Due to **power limitations**, mobile units

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- Due to **power limitations**, mobile units transmitted not directly to the BS but to **receiving sites**

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- How a call happen in MTS ?
 - The caller first called a **special number** to connect to an **MTS operator**.
 - The **caller informed the operator** of the **mobile subscriber's number**.
 - Then the operator searched for an idle channel in order to relay the call to the mobile terminal.

Mobile Telephone System (MTS)...

- Major limitations:
 - Manually Switching of calls.
 - limited channels are available (3 channels)
- Improved Mobile Telephone System (IMTS)
 - Utilized *automatic call switching*
 - Enhanced its mode of transmission → *Full-duplex*
 - The *intermediation of the operator* in a call Eliminated.
 - Number of channel increased to 23
 - IMTS has **problem**:
 - providing a small capacity Because spectrum usage.
 - Interference to adjacent systems due to *large power of BS transmitter*.

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Check detail of 1G ? ...

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- Second Generation (2G)
 - Second-generation wireless/mobile communication technology.
 - The first digital cellular networks $\mapsto$ Digital modulation
 - Highly employ **TDMA** and **CDMA** techniques to assign users.
 - Has brought :
 - Digital traffic channels
 - Encryption
 - 2G systems provide encryption to prevent eavesdropping.
 - Error detection and correction
 - digital traffic allows for detection and correction, giving clear voice reception.
 - Channel access : dynamically shared by a number of users.
 - various standards 2G standards $\mapsto$ GSM, IS-136, PDC, CDMA/IS-95
 - The most notable 2G standard system



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Global System for Mobile Communication(GSM)

GSM Systems

- GSM completely designed from scratch $\mapsto$ backward Incompatible with 1G
- It can deliver data rate up to 9.6 Kbps by using 124 channels per cell;
- Each channel can support 8 users through TDMA
- GSM has defined three different categories of services:

1. Bearer Services

- services for data transmission
- 9600 bit/s $\mapsto$ non-voice data
- those services provided to transport binary data across the GSM network.
- Operate in 2-Mode
 - Transparent Bearer services
 - only use the functions of the physical layer to transmit data
 - Data transmission has a constant delay and throughput
 - forward error correction (FEC)
 - Non-transparent Bearer services
 - has Radio Link Protocol (RLP) on top of transparent bearer services
 - Employ protocols in Link layer and network layer for reliability
- defines more services $\mapsto$ for Interworking with PSTN, ISDN and packet switched public data networks (PSPDN)

2. Tele Services

3. Supplementary services.

- GSM Systems ...

- 2 Tele Services

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 - Emergency number
 - mandatory and free of charge
 - Short message service (SMS)
 - limited upto 160 characters
 - Enhanced message service (EMS)
 - larger message size, formatted text, and the transmission of animated pictures, small images
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 - Has brought :
 - User identification
 - Call redirection
 - Forwarding of ongoing calls
 - barring of incoming/outgoing calls

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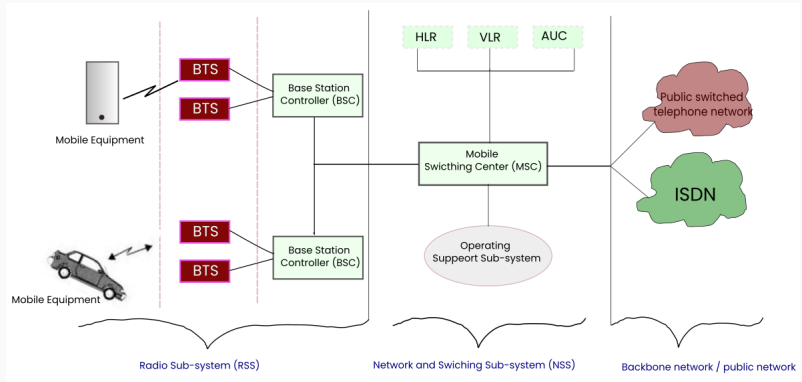


Figure 1: GSM Architecture

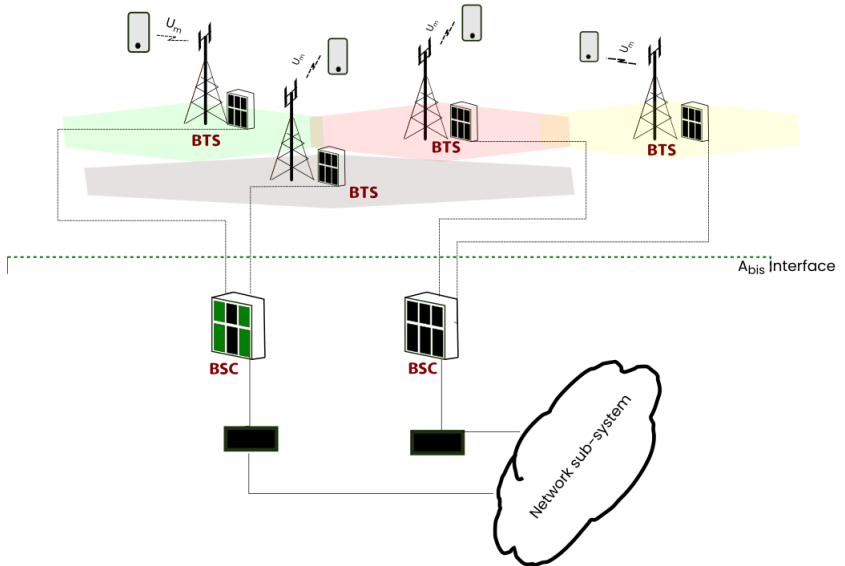
CH01: Fundamentals of Mobile Computing

- GSM composed of 3 Sub-systems :
 - Radio sub-system
 - Network and Switching subsystem
 - Operating subsystem

Radio Sub-system

- comprises all radio specific entities, $\mapsto$ i.e. Mobile nodes and BSS
 - Mobile nodes $\rightarrow$ You Know
 - Base Station subsystem (BSS)
 - Consist of many BSSs $\mapsto$ controlled by BSC.
 - Responsible for :
 - Maintain connection to Mobile Node.
 - Coding and decoding voice
 - Base station controllers (BSC)
 - provides all the control functions and physical links between the MSC and BTS
 - High-capacity switch that provides function.
 - Handover,
 - Cell configuration data
 - Control of radio frequency (RF) power levels
 - Base transceiver station (BTS):
 - is the radio equipment (transceivers and antennas) needed to service each cell in the network.
 - A group of BTS's are controlled by an BSC.

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Network Switching Subsystem

- Responsible for performing **call processing** and **subscriber related functions**
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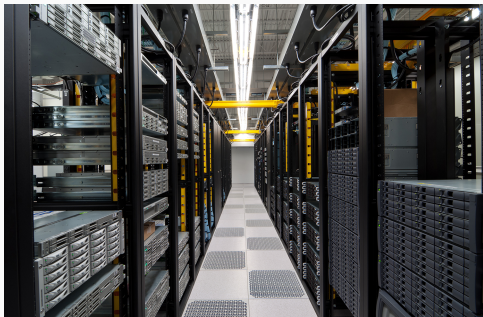
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 - It is a **database** that contains information about the identity of mobile equipment that prevents calls from stolen, unauthorized or defective mobile stations

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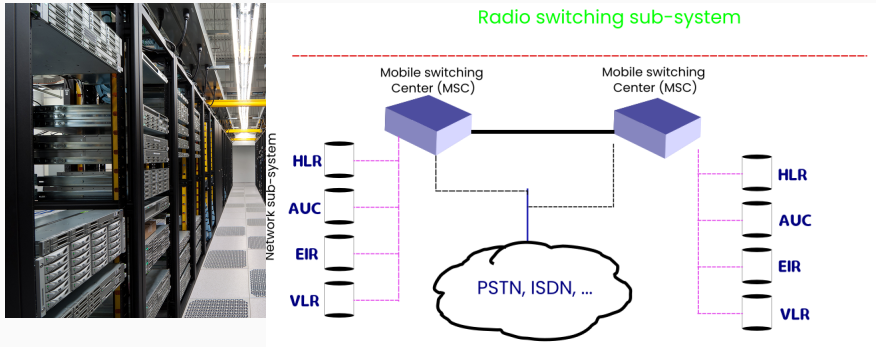


Figure: Network switching sub-system (NSS)

- The operation subsystem (OSS)



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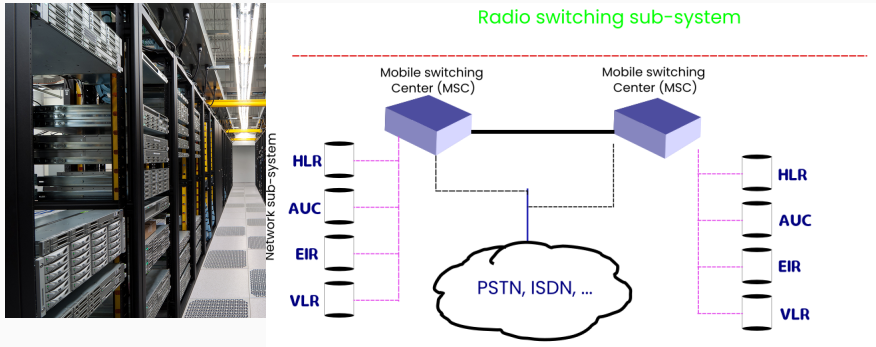


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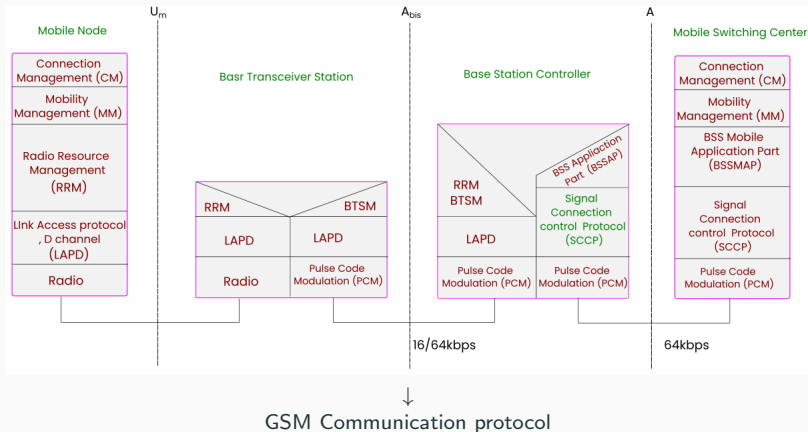


Check what is the key tasks ? ... read it

- How Components of GSM communicate ?
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2. Link Layer

- LAPD_m protocol has been defined at the U_m interface for layer two.

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 - **Lower sub-layer**: Radio resource management (RRM).



Setup, maintenance, and release of radio channels.

- **Upper sub-layer** : Mobility management (MM)



registration, authentication, identification, location updating,
and the provision of a temporary mobile subscriber identity (TMSI).

4. Cell Mgmt Layer

- Second and a half-generation (2.5G) Mobile Communication system



is used to describe 2G-systems that have implemented a

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is used to describe 2G-systems that have implemented a packet switched domain in addition to the circuit switched domain



- has actually brought:
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- GPRS ⇨ Stone features

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High-speed wireless Internet

- Other data communications



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data transmission rates of 28 kbps and higher
- EDGE ⇨ Stone features
 - increases transmission speeds on GSM networks and
 - enables the transmission of large amounts of data at 384Kbps

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- In addition, Both support:
 - WAP
 - MMS
 - SMS
 - mobile games
 - data transmission rates of 144 kbps or higher,
 - etc



- The third generation (3G Mobile system)
 - Just provides high-speed data transmissions of 144kbps and higher
 - 3G wireless networks support the following maximum data transfer rates:
 - 2.05 Mbits/second $\mapsto$ stationary devices.
 - 384 Kbits/second $\mapsto$ slowly moving devices, such as a handset carried by a walking user.
 - 128 Kbits/second $\mapsto$ fast moving devices, such as handsets in moving vehicles.
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 - Comes with enhanced features
 - higher data speed.
 - Multiple mobile applications and clarity of digital signals.
 - Enhanced audio and video streaming;
 - Video-conferencing support;
 - Web and WAP browsing at higher speeds;
 - IPTV (TV through the Internet) support
- Is there a 3G Standard?

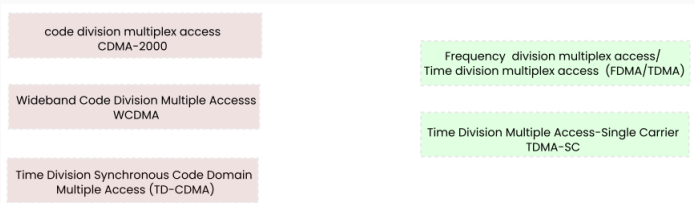


International Telecommunication Union (ITU) is responsible for standardizing 3G

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based on **CDMA** Technology

Based on **TDMA** Technology

- The well-known **Universal Mobile Telecommunications System (UMTS)**



W-CDMA operating at 2.1GHz ⇨ most of the world's wireless operators have chosen to use UMTS.

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- Later 3G releases, often denoted 3.5G and 3.75G,



provide mobile broadband speeds of several Mbps to $\mapsto$ smartphones, tablets and mobile modems in laptop computers

- **HSUPA (High-Speed Uplink Packet Access)**



upgrades the uplink for higher data transfer speed and capacity of 1.4, 1.9, 5.8Mbps.

- **HSDPA High-Speed Downlink Packet Access**



having faster speed from 1.8, 3.6, 7.2 to 14Mbps.

- **3.75G** $\mapsto$ HSPA High Speed Packet Access
 - is a combination of **HSUPA** and **HSDPA** that improves the performance of **W-CDMA** and can attain speed 21 Mbps to 28 Mbps.
 - **HSDPA High-Speed Downlink Packet Access** having faster speed from 1.8 ,3.6 ,7.2 to 14 Mbps.

- Was there anything though (3G) ?

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 - High Bandwidth Requirement $\mapsto$ high performance application
 - High spectrum Licensng
 - Huge Capital
 - Need of Mobiblity and Service Portability
 - The Need of wider bandwidth



Motivated towards 4G

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The other Contribution of ITU $\mapsto$ Fourth Generation mobile system (4G)



specifies the key characteristics of the standard, (such as transmission technology and data speeds.

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- 4G $\mapsto$ refers to all-IP packet-switched networks, mobile ultra-broadband access and multi-carrier transmission
- With 4G download speeds $\mapsto$ wireless users can stream high-definition video and audio.
- X-tics of 4G Mobile System
 - 4G communication provides 100 Mbps for mobile users and up to 1 Gbps over fixed stations.
 - is an Adoption of Packet Switching instead of Circuit-Switching in voice and video calls
 - will also be able to stream HD television and radio hand-held devices
- Any Tech. behind 4G ?
 - Smart Antennas for Multiple-Input and Multiple-Output
 - Orthogonal Frequency Division Multiplexing (OFDM)
 - IPV6
 - VoIP
 - Software defined Radio (SDR) System

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- Features of 4G
 - Fully IP-Based Mobile system
 - High speed, High capacity and low-cost per bit
 - Global access, service portability
 - Seamless switching and variety of QoS
 - Ad-Hoc and Multi-hop network
 - Better spectral efficiency
 - Interoperability with existing wireless standards
- **LTE (Long-Term Evolution)**
 - Standard managed by 3GPP (3rd) Generation Partnership Project Organization
 - designed to provide up to 10 times the speed of 3G Network for mobile devices
 - LTE is standard along Ultra Mobile Broadband and WiMAX (IEEE 802.16)
 - LTE has a theoretical net bit rate
 - 100 MBit/S, ↓ speed
 - 50 MBit/S ↑ Speed

↓

if a 20 MHz channel is used and more if multiple-input multiple-output (MIMO), i.e. antenna arrays, are used.

Mobile Cloud Computing



- Why Mobile Cloud Computing ?



Think The Scarcity in Mobile Devices $\mapsto$ Such as battery problem, storage and bandwidth

- Mobile Cloud Computing offers advantages to users by allowing them to utilize infrastructure, platforms and software by cloud providers.

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- Mobile Cloud Computing
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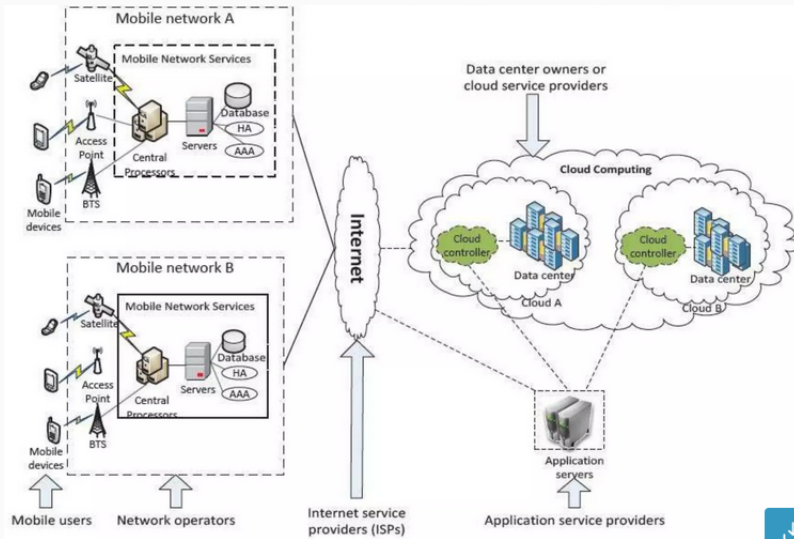
- Mobile Cloud Computing offers advantages to users by allowing them to utilize infrastructure, platforms and software by cloud providers.

Defination

- Cloud Computing:
 - is storing and Accessing data and programs via TCP/IP network instead in device's storage devices
- Mobile Cloud Computing
 - refers to an infrastructure where both the data storage and processing happens outside the mobile device.



CH02: Mobile Communication Technologies



♣ Do we have driving forces (enabling technologies) ? $\mapsto$

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- Enabling Technologies

- Communication system (3G, 3.5G, 3.75G, 4G, 5G)
- HTML, CSS
- Hypervisor
- Web-services

♣ Any Merits from Mobile cloud computing ? $\mapsto$

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♣ Any Merits from Mobile cloud computing ? ⇨ Yeah!

- Merits

- Flexible and portable
- On-demand services
- Cost effectiveness
- Universal access
- allows to make backup and recovery
- centralized monitoring and maintenance of software

♣ Any demerits from Mobile cloud computing ? $\mapsto$

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- Disadvantages of Mobile Cloud computing

- Trust, Security and privacy issues
- Data Security
- connectivity and performance issues
- Power consumption
- Congestion issues.

♣ Appication of Mobile Cloud Computing $\mapsto$

♣ Any demerits from Mobile cloud computing ? ⇨ Ofcourse!

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♣ Application of Mobile Cloud Computing ⇨ Yeah!

- Mobile Commerce
- Cloud Email
- Mobile Learning
- Mobile Healthcare
- Cloud Music
-

Done!
Question!